

A PROJECT TITLE

ON

**Breast Cancer Classification Using Decision Trees and Random Forest with
ROC-AUC/PR-AUC Evaluation**

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Name of Title: Breast Cancer Classification Using Decision Trees and Random Forest with ROC-AUC/PR-AUC Evaluation

1. Abstract:

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. Early and accurate diagnosis plays a crucial role in improving treatment outcomes and increasing patient survival rates. Machine learning techniques have emerged as effective tools for assisting healthcare professionals by analyzing medical data and identifying patterns that may not be easily recognized through traditional methods. This project presents a comparative study of two widely used machine learning algorithms, Decision Tree and Random Forest, for the classification of breast cancer tumors as benign or malignant using the Breast Cancer Wisconsin Diagnostic dataset.

The project begins with data preprocessing, including data exploration, handling missing values, and splitting the dataset into training and testing sets. Both models are trained and evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, confusion matrix, Receiver Operating Characteristic (ROC) curve, Area Under the ROC Curve (ROC-AUC), Precision-Recall (PR) curve, and PR-AUC. Cross-validation and hyperparameter tuning are also performed to improve model reliability and generalization. Additionally, feature importance analysis is conducted using the Random Forest algorithm to identify the most influential features contributing to the classification process, thereby improving the interpretability of the model.

The experimental results demonstrate that the Random Forest classifier generally achieves higher predictive performance and better robustness than the Decision Tree classifier due to its ensemble learning approach, which reduces overfitting and improves generalization. The findings highlight the effectiveness of machine learning in supporting early breast cancer diagnosis while emphasizing the importance of selecting appropriate evaluation metrics for medical classification problems. This project provides a comprehensive understanding of classification algorithms, model evaluation techniques, and explainable machine learning, making it a valuable learning experience for beginners and a practical foundation for developing intelligent healthcare decision-support systems.

2. INTRODUCTION:

Breast cancer is one of the leading causes of cancer-related deaths among women worldwide. According to global health studies, early detection and accurate diagnosis significantly improve treatment outcomes and increase the chances of survival. Traditionally, breast cancer diagnosis relies on clinical examinations, medical imaging, and laboratory tests. However, the increasing availability of medical data has created opportunities to apply machine learning techniques to assist healthcare professionals in making faster and more accurate diagnostic decisions.

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from historical data and make predictions without being explicitly programmed for every scenario. In the healthcare sector, machine learning has been widely adopted for disease prediction, medical image analysis, patient risk assessment, and decision support systems. By analyzing patient data, machine learning models can identify complex relationships among medical features, helping improve diagnostic accuracy and reduce human error.

This project focuses on the classification of breast cancer tumors into two categories: benign (non-cancerous) and malignant (cancerous). The study uses the Breast Cancer Wisconsin Diagnostic dataset and applies two supervised machine learning algorithms: Decision Tree and Random Forest. Decision Trees provide a simple and interpretable classification model by creating decision rules based on feature values, while Random Forest improves prediction performance by combining multiple decision trees through an ensemble learning approach.

The performance of both models is evaluated using various classification metrics, including accuracy, precision, recall, F1-score, confusion matrix, Receiver Operating Characteristic (ROC) curve, Area Under the ROC Curve (ROC-AUC), Precision-Recall (PR) curve, and PR-AUC. In addition, feature importance analysis is performed to identify the most influential attributes contributing to breast cancer prediction.

The primary objective of this project is to compare the effectiveness of Decision Tree and Random Forest algorithms in breast cancer classification and to determine the model that provides better predictive performance. The project also aims to provide a practical understanding of the complete machine learning workflow, including data preprocessing, model training, evaluation, and interpretation. The findings demonstrate how machine learning can support early breast cancer diagnosis and serve as a valuable decision-support tool in modern healthcare.

3. SOFTWARE REQUIREMENTS:

The following software and tools are required to develop, train, evaluate, and visualize the machine learning models used in this project.

Software/Tool	Version (Recommended)	Purpose
Google Colaboratory (Google Colab)	Python 3.12	Cloud-based development environment for writing and executing Python code.
Python	3.10 or above	Primary programming language used for implementing machine learning algorithms.
NumPy	2.5.2	Numerical computations and array operations.
Pandas	3.0.5	Data loading, manipulation, and preprocessing.
Matplotlib	3.11.1	Data visualization and plotting graphs.
Seaborn	v0.13.2	Statistical data visualization and correlation analysis.
Scikit-learn	1.9.0	Implementation of machine learning algorithms, model evaluation, preprocessing, and cross-validation.

Signature of the Guide

Course Instructor